ASSESSMENT TOOL – 3

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Course Code: CSA1511

Course Name: Cloud Computing

1. Define cloud computing and explain any four characteristics that make it different from traditional on-premise computing.

A) Define Cloud Computing and Explain Four

Characteristics Definition:

Cloud computing is the delivery of computing services (servers, storage, databases, networking, and software) over the Internet on demand.

Characteristics:

- On-demand self-service: Users access resources whenever needed.
- Broad network access: Services are available over the Internet.
- Resource pooling: Resources are shared among multiple users.

Scalability & Elasticity: Resources can be increased or decreased based on demand.

2.Trace the evolution of cloud computing from hardware evolution to internet software evolution, and show how these changes enabled modern cloud platforms.

A) Evolution of Cloud Computing

- Hardware Evolution: Powerful servers and virtualization enabled efficient resource sharing.
- Internet Evolution: High-speed Internet made remote access possible.
- Software Evolution: Web applications and APIs enabled online services.

- Modern Cloud: Combines virtualization, automation, and Internet technologies to provide scalable cloud platforms.

3.Explain the role of server virtualization in cloud computing. Differentiate between virtualization as an enabling technology and cloud computing as a service model.

A) Role of Server Virtualization

Server Virtualization:
Allows multiple virtual machines (VMs) to run on a single physical server, improving resource utilization.

Difference:

Virtualization	Cloud Computing
Technology	Service model
Creates virtual machines	Delivers services over the Internet
Focuses on hardware efficiency	Focuses on providing on-demand resources

4.Discuss the role of data centers in delivering cloud services.

A) Role of Data Centers in Cloud Services

- Store servers and networking equipment.
- Host cloud applications and data.
- Ensure security and backups.
- Provide high availability and reliable performance.

5. Differentiate IaaS, PaaS, SaaS, and XaaS with one suitable real-world example for each model.

A)

Model	**Purpose**	**Example**
IaaS	Provides virtual infrastructure	Amazon EC2
PaaS	Provides development platform	Google App Engine
SaaS	Provides ready-to-use software	Google Workspace
XaaS	Any service delivered via cloud	AWS (Storage, Database, AI services)